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Review

Addressing context dependence in studies of plant diversity to improve the understanding of natural enemy conservation

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Increasing plant diversity has become a major aspect of habitat management and natural enemy conservation. Nonetheless, the results of plant diversity studies have varied significantly within and across agroecosystems. This variation has often been ascribed to a condition known as context dependence. However, concluding plainly that results are context dependent does not allow for the understanding of the actual underlying causes. Therefore, I discuss in this paper the importance of identifying and dealing with context dependence. I specifically comment on common biotic and abiotic inherent variables that can drive context dependence. The most common context types explored herein are location, time, nonfocal plant and crop species, and natural enemy species. Finally, I offer several recommendations for identifying and dealing with context dependence. I believe understanding the different forms in which context dependence arises is paramount to reduce unexplained variation and improve the predictability of plant diversity studies.

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Introduction

It is widely known that plants can directly and indirectly influence the diversity/abundance of arthropods in a given habitat [1]. Most often, this influence is driven by morphological and/or nutritional plant traits that could either benefit or hamper specific arthropod species [2,3]. In the perspective of agriculture, entomologists and plant scientists have long strived to underpin the mechanisms through which plants can mediate interactions between pests and their natural enemies (predators and parasitoids). Likewise, biological control practitioners are interested in how plants can promote the abundance/ diversity of natural enemies in a manner to engender pest suppression. In this respect, it is already known that plant diversity has the potential to promote natural enemies through a phenomenon known as associational effect [4]. Specifically, associational effect is a type of indirect interaction that occurs between neighbor plants [6,7], where one of the possible outcomes includes a neighboring plant (hereafter also referred to as nonfocal plant) facilitating the biological control of pests on the nearby focal plant [4]. These nonfocal plants can consist of either agronomic or nonagronomic plant species, and they could occur naturally or be purposefully cultivated. In the context of this review, the term nonfocal plant can refer to a companion plant, an intercrop, or weed species (herein considered beneficial to natural enemies), which occurs nearby focal plant species (e.g. crop). Furthermore, I choose to use in this review the term plant diversity (to refer to a number of plant species) instead of plant richness because the former is more widely used in the literature related to conservation biological control (CBC). CBC can be defined as the "modification of the environment or existing practices (e.g. favor the use of selective insecticides) to protect and enhance specific natural enemies or other organisms to reduce the effect of pests" [5]. Modification of the environment is also interchangeable with the term habitat management (HM), which concerns the manipulation of the agricultural environment (e.g. plant diversity) to maximize the function of biological control. In this respect, nonfocal plants can often promote biological control of pests on focal plants by providing natural enemies with food to boost fitness (pollen, nectar, alternative prey/host, derived honeydew) and/or nonfood resources to conserve them (shelter, microclimate, mating site, refuge from adversities) [8,9]. As CBC-HM emerges as a promising approach to curtail the excessive use of insecticides/ acaricides and foster sustainable agriculture, a plethora of studies concerning the promotion of nonfocal plants in the field (i.e. increasing plant diversity) has been amounting to an extensive literature. Nonetheless, the outcomes of such studies have varied significantly within and across agroecosystems, resulting in neutral, negative, or positive effects [8,10], which in turn have been challenging entomologists as well as hindering the adoption of CBC-HM by growers.

The discrepancy in results from plant diversity studies has increasingly been ascribed to a condition known as context dependence. In short, context dependence serves to denote circumstances where ecological relationships vary according to the conditions (i.e. the context) under which they take place. This term, context dependence, has become a common form to describe results variation within and between studies that investigate the same question/process [11]. However, by and large, concluding that results are context dependent does not allow for the understanding of the actual underlying causes of the varying results. Furthermore, the perfunctory and widespread use of the term context dependence may mislead people to think that CBC-HM is mostly unpredictable and that the diverging results can only make sense in a case-by-case basis. A typology of context dependence in ecology has been put forth by Catford et al. [11] who identified two forms of context dependence that may arise from four potential sources (see below i–iv), which can also be operating on CBC-HM (i.e. usage of nonfocal plants to promote natural enemies). First, there is the mechanistic context dependence where a relationship between variables (e.g. plant diversity and natural enemy abundance) differs under different spatiotemporal/ecological conditions due to (i) interactions with a third variable (e.g. climate) that modifies that relationship. This could be an ecologically realistic context dependence, and therefore, it could be predictable if the proper variables are understood and measured. Second, there is the apparent context dependence in which the relationship between variables (e.g. plant diversity and natural enemy abundance) does not actually differ under distinct contexts, but sometimes appear to due to (ii) the presence of confounding effects, (iii) pitfalls in statistical inference and statistical power, and/or to (iv) differences in the methodology used among studies. The sources of apparent context dependence (i.e. confounding factors, statistical inference, and methodological differences) might be more easily preventable by carefully planning ahead both the experimental design and the methodology for data collection. In the case of mechanistic context dependence, it is essential that in addition to planning carefully the experimental design and methodology, we also inspect the data appropriately and rigorously for interaction effects and potentially overlooked variables as well as prioritize whenever possible smaller-scale/in-depth studies over superficial/broader ones [11]. Moreover, following ideas put forth by Heimpel and Jervis [12], one could also reduce chances of mechanistic context dependence and improve predictability by establishing a stepwise (tiered) set of premises to determine whether or not a specific nonfocal plant species could really be relevant for the success of CBC-HM in the field (see recommendations).

In light of the necessity to improve the predictability and mechanistic understanding of CBC-HM implementation, I henceforth delineate in this paper a discussion around what biotic and abiotic variables could lead studies of plant diversity to produce mechanistic context-dependent results. More specifically, I explore inherent variables linked to each common context type (i.e. location, time, plant, and natural enemy species) that could be conducive to mechanistic context-dependent results (Table 1). Furthermore, I present potential general approaches that could help preventing and/or dealing with context-dependent data from such studies. The most common context types explored in this paper will be location, time, plant species (nonfocal and focal plants) and natural enemy species, which can directly and indirectly affect the abundance and quality of plants (as food or nonfood resource for natural enemies). Nonetheless, I do not base my arguments and discussion on an extensive review of the literature as there is not yet specific published studies addressing this topic in CBC; but instead, I use key papers to explore the most common potential variables that could cause context dependence in plant diversity studies. Additionally, the

Table 1

Sample of context types and their inherent variables, which are likely to occur in nonfocal plants studies. '*Interactions*' imply all potential relationships between two or more of the inherent variables within context type.

Sample of context types	Location	Time	Nonfocal plant species	Crop species	Natural enemy species
Inherent variables	Temperature Light Humidity Soil Management Landscape Interactions	Temperature Light Humidity Soil Management Landscape Interactions	Spatial arrangement Genotype Phenotype Alternative prey/host Interactions	Spatial arrangement Genotype Phenotype Pest species Interactions	Diet breadth Body size Behavior Movement type Phenology Interactions

main goal of this paper is to communicate the importance of addressing context dependence properly in CBC-HM studies and not to explore exhaustively all the potential variables that could lead results to context dependence. Finally, although these context types are presented separately under subheadings (see below) to facilitate organizing and reading, they are still interdependent in many aspects and therefore could often interact with each other. For example, it would be difficult to discuss landscape scale context without considering interactions with variables, such as species dispersal capacity, resource requirements, or life history strategies.

The context of location and time

The effects of location (space) and time are always interrelated, and in this case, the factor time will frequently influence variables that are inherently linked to location. This happens because location/space variables are often prone to temporal changes. Regardless, the most common variables affected by both contexts of location and time are temperature, light, humidity, soil quality, farm management, landscape structure/composition, and the potential interactions among themselves (Table 1), all of which can influence nonfocal plants in the field. In fact, it has been reviewed that several environmental variables, such as water availability, temperature, ambient humidity, soil factors, and light, can affect nectar abundance and quality on flowering plants [13], which, in turn, can be expected to influence omnivorous natural enemies. Likewise, the production and usefulness of extrafloral nectar for CBC-HM have also been shown to depend on ecological contexts [14].

In terms of landscape structure/composition, it has become increasingly more apparent that the context of location is a very common source of results variation when CBC-HM studies are replicated in space [15,16,8]. When natural enemies are supported by the wider landscape, some of the variation in results may be due to differential responses from distinct groups of natural enemies to landscape structure. Furthermore, natural enemy responses may be influenced by patches of nonfocal plants interacting with landscape structure through mechanisms, such as complementarity, source-sink relationships, meta-populations, and/or neighborhood effects [17,18]. Nevertheless, previous studies have reported that even in simplified landscapes (poor quality), nonfocal plants may still support natural enemy populations and thus show potential to release, or at least relax, this spatial context dependency. For example, Woltz et al. [19] found out that landscape complexity and nonfocal plants (insectary plants) had independent positive effects on the abundance of coccinellids, which caused a negative impact on the population of soybean aphids. Regardless, it is still rarely investigated whether specific trait(s) of nonfocal plants are affected by location context and how this would play out in influencing CBC-HM implementation. Therefore, more investigations are needed in that particular realm.

In previous studies, our research group has kept experimental design/methodology and location constant while changing the timing (i.e. seasons) of companion plant experiments (see below). It is noteworthy that not all response variables respond the same way to the context of time. In that sense, some variables will respond differently with seasons, whereas others will respond the same way across time. For example, in a two-season experiment with brassicas (summer and winter), the companion plants alyssum flowers *Lobularia maritima* (L.) were able to engender faster aphid suppression only during the winter time, whereas in the summer, there was no difference between treatments [20]. In contrast, in the same experiment, when we looked at the influence of companion plants on brassica production, we found that plant fresh mass was constantly higher in the treatment with companion plants, regardless of time (season). Likewise, when we investigated the influence of intercrop plants (brachiaria) on maize pests, we found that pests such as the leafhopper *Dalbulus maidis* (DeLong and Wolcott) would be constantly higher in the intercrop treatment in all seasons, whereas the abundance of the leaf beetle *Diabrotica speciosa* (Germar, 1824) would depend in the context of time [21]. Although these studies are informative and instrumental for entomologists and growers, further studies are required to disentangle the temporal relationship of pest regulation and habitat management (use of nonfocal plants). That is why when replicating experiments in space and time it is also paramount to identity and measure most of the variables that are closely linked to location and time, which in turn could drive mechanistic context dependency (Table 1). In addition, taking measurements on key response variables from nonfocal plants (e.g. nectar and pollen quality) would be also important to inspect for potential underlying causal relationships (e.g. increased or reduced abundance of natural enemies in the presence of nonfocal plants). Finally, understanding the spatiotemporal movement of natural enemies between nonfocal and focal plants within the landscape is key for dealing with context dependence associated with location and time. For example, events of natural enemy immigration and emigration are important in determining the population dynamics and pest suppression in the field [36].

The context of nonfocal and focal plant species

The results of CBC studies will also likely vary according to the plant species involved in the investigation. In this particular case, both the species of nonfocal and focal plant (i.e. crop) will influence directly and/or indirectly the recruitment and establishment of natural enemies in a given area and time. The most common variables that can be associated with the nonfocal or focal plant species and influence CBC-HM include the spatial arrangement of plants (at local scale), genotype, phenotype, and the potential interactions among themselves. In addition, the pest species and alternative prey/host species occurring, respectively, on the focal and nonfocal plants will also contribute to variations in results of CBC-HM studies (i.e. context dependence). The aforementioned variables should therefore be identified and considered *a priori* in experimental designs concerning companion plant studies to help preventing and dealing with mechanistic context dependence (i.e. context of plant species).

The spatial arrangement of nonfocal plants and/or crop can influence the dynamics of pest and natural enemies [22–24]. The spatial arrangement of nonfocal plants can often modulate the searching behavior of natural enemies in a manner that could either favor or hamper biological control. For example, Jaworski et al. [22] found that when companion plants were distributed more evenly within apple orchards, they recruited predators more quickly, which in turn mediated a faster aphid suppression. In regard to crop spatial arrangement, Rakotomalala et al. [25] found that row and strip are more effective than mixed and relay intercropping in promoting natural enemies and suppressing pests, which in turn also offers an opportunity for mechanistic context dependence.

The pest species and/or alternative prey/host species occurring on the focal or nonfocal plants may also engender context dependence as various of their attributes are prone to variation (e.g. size, nutritional quality, and abundance) in response to extrinsic factors, which in turn could ultimately influence natural enemies. Among the various possibilities for plant species to influence pest control, the focal and nonfocal plant species can often affect the quality and abundance of the pest and/or prey/host in a manner that it will influence natural enemies through a cascade effect [26]. For example, Peñalver-Cruz et al. [27] found that the parasitoids *Aphelinus mali* originating from aphids *Eriosoma lanigerum* reared on the companion/banker plant *Pyracantha coccinea* were able to parasitize more aphids than those parasitoids originating from *E. lanigerum* reared on apple plants. In addition, the focal and nonfocal plant species may also drive the honeydew quality and quantity in a manner that it can indirectly affect the recruitment and establishment of natural enemies (e.g. parasitoids). For example, Monticelli et al. [28] demonstrated in their study that the host plant species affects significantly the fructose content of aphid honeydew, which influenced parasitoid nutrition and survival. Finally, the abundance of pest and/or alternative prey/host may also sometimes produce context dependence by functioning as a confounding factor. In this particular case, the natural enemy response will often be density dependent, thereby making it necessary to at least consider pest abundance as a covariable in the analysis.

The context of natural enemy species

Natural enemy species or functional group is also a source of context dependence as often their response to nonfocal plants, and the subsequent biological control will be influenced by traits that are inherent to the biological control agent species/functional group. In other words, the response of natural enemies to plant resource use is almost always group or species specific. Some of the most common traits from natural enemies that will affect CBC-HM outcome and potentially engender context dependence include diet breadth (specialist vs generalist), body size, behavior, phenology, and movement type (flying vs epigeal) [2,29]. Understanding the context dependence engendered by natural enemy diversity [30] is paramount for predicting the success of multiple biological control agents and informing whether conservation of natural enemy diversity and biological control are congruent goals. In that sense, it has been theorized that pairs of response-and-effect traits involving nonfocal plants and natural enemies should be identified/selected to improve resource provisioning to biological control agents and potentially help making better sense of context-dependent results in CBC-HM studies [31,32].

When analyzing CBC-HM data, it is important to look at the responses of specific natural enemy species or functional groups to inspect for context dependence. For example, it has been shown before that specialist and generalist natural enemies will have different responses to plant diversity, where specialists are less efficient than generalists in diversified agroecosystems [33,34]. Likewise, Carvalho et al. [35] showed in their study that while flowers from companion plants *L. maritima* mediated a faster recruitment of specialists (parasitoids), the visual cues from the focal host plant (brassicas) were exploited more quickly by flying predators as opposed to epigeal ones. Therefore, it would be a good analytical practice to look at the data separately by natural enemy species or functional groups, which then would likely enhance the understanding of apparently complex results. Furthermore, inspecting for potential statistical interactions would be an appropriate way to analyze these types of data. For example, a statistical model examining the response of generalists versus specialists abundance to plant diversity could look like the following model: Abundance ~ PlantDiversity + FeedingGuild + PlantDiversity*FeedingGuild, where '*' stands for interaction.

Final remarks and general recommendations

Because CBC-HM studies usually involve multiple trophic levels and need to be appropriately replicated in time and space, the chances for context dependence increase substantially. Although the sources of apparent context dependence should always be considered and minimized, it is the mechanistic context dependence that is more likely to happen in CBC-HM studies and thus needs greater attention. As seen in this short review, the possibilities for interaction effects arising from plant diversity studies are vast, and thus, so are the chances for mechanistic context dependence. To better control for mechanistic context dependence and to improve its understanding, I recommend the following: (i) identify and take measurements on all potential key covariables, (ii) use nested designs and mixed effects analysis whenever appropriate (specially to control for random effects and temporal variability), (iii) consider the use of piecewise structural equation modeling (PiecewiseSEM) when dealing with multiple hypotheses and/or when interested in investigating direct and indirect effects of nonfocal plants on natural enemies, (iv) examine the data carefully and rigorously for interactions and potentially overlooked variables, (v) prioritize smaller and more-in-depth studies when appropriate, (vi) collect data on multiple response variables, and analyze them together as well as separately (e.g. by species or functional group), (vii) collect data using a functional trait approach for plants and arthropods (e.g. response-and-effect traits) when appropriate, and (viii) prioritize manipulative experiments over observational ones whenever possible. Furthermore, we could reduce chances of context dependence as well as improve predictability by examining and demonstrating in a stepwise manner that (1) natural enemies are food/refuge limited in the area we intend to implement CBC-HM, (2) this limitation can significantly be alleviated by complementing food/refuge via provision of specific nonfocal plant species, (3) an ensuing increase in natural enemy survival would arise, (4) an enhanced population (abundance) of natural enemies would also take place, and finally (5) all these sequential events would culminate in an improved predation/parasitism of the pest. Despite the apparent complexity of this theme, researchers should always take pre-emptive measures to account for potential context dependence in their studies. Taken altogether, I believe that recognizing and addressing properly cases of context dependence will be paramount to improve the predictability of CBC-HM and hence increase its adoption in agriculture.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The author declares that he has no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

The author declares that there was no use of AI and AI-assisted technologies in the writing process of the current article.

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